

Yuhao Yuan

Seeking PhD Opportunities for Fall 2027

Research Interests: AI in Education, Educational Data Mining, and Learning Analytics

✉ yuhao.yuan@mail.utoronto.com 📞 +1 582 203 8954 📍 Toronto 🌐 <https://yyhaos.github.io/>

EDUCATION

University of Toronto

Sept 2025 – Now

Master of Education, Ends in Jul 2027

Beijing Normal University

Sept 2016 – Jul 2021

Bachelor of Computer Science

WORK EXPERIENCE

Software Developer - ByteDance Ltd. China

June 2021 - Jul 2024

- Supported Continuous Integration and Continuous Delivery (CI/CD) of core components for the TikTok China iOS application, ensuring reliable and timely releases.
- Established and maintained a comprehensive automated testing framework—including unit, UI, interaction and integration tests, which covering automated test template generation, coverage merging and trucking, and git hooks validation workflows.
- Built and maintained an automated release management system covering version publishing, standardized commit validation, and semantic versioning, streamlining release cycles and reducing manual errors.
- Reverse-engineered the SwiftUI Text component and contributed to the development of the internal DanceUI library, enhancing text rendering performance and extending UI functionality.

FrontEnd Software Developer Internship - Alibaba Ltd. China

Jun 2020 - Sept 2020

- Used SIMD (Single Instruction, Multiple Data) to apply a blur effect and smooth out the pictures in Rust.

ACADEMIC EXPERIENCE

Software Engineer - pyKeyATM project

Jan 2025 - Now

- Collaborated with Dr. Ji at the University of Toronto to Develop the pyKeyATM R package, designed for abstracting topic words from documents.

Research Software Engineer - rBKT Project

Aug 2024 - Dec 2025

- Developed an open-source R implementation of Bayesian Knowledge Tracing for estimating student knowledge mastery from problem-solving sequences.
- Contributed to model training and prediction APIs, data validation, simulation, evaluation, parameter-recovery experiments, and package documentation.

Software Engineer - Entrepreneurial Project

Aug 2021 - Jul 2024

- Developed an Android App called “keysManager” using KotlinLanguage
- Launch the product outside of school, continuously tracked the users’ online activities, and updated the App according to the target market’s feedback
- Accomplished the goal of the App to manage user’s passwords,create secure passwords, set up network Keychain, and automatically sync all data while maintainingnetwork security

PUBLICATIONS

- Yuan, Y., Zhou, B., Qi, J., Luo, N., & Ji, F. (2026). BKT: A Bayesian knowledge tracing package for the R environment. *Behavior Research Methods*, 58(5), Article 134. doi:10.3758/s13428-026-02955-9